

POLI110-Blocking Confounding with Experiments:

An Expanded Overview of Conditioning-Based and Design-Based Approaches

Aodi Cheng

Abstract

This pdf explains why experiments can effectively block confounders, making it easier to identify causal effects. It introduces the potential outcomes framework, illustrates the role of randomization, and compares *conditioning-based* methods (e.g., regression, matching) with *design-based* methods (e.g., randomized controlled trials, Difference-in-Differences). A simple Directed Acyclic Graph (DAG) demonstrates how confounding arises, while a table summarizes the key differences between these two broad classes of approaches. Relevant formulas are displayed separately to maintain clarity in the exposition. Compared to earlier versions, this document expands on the strengths, limitations, and typical use cases for each approach, and illustrates these concepts through a detailed analysis of four research designs examining running shoe performance.

Introduction

Researchers in field of political science often wish to determine whether a variable X has a causal impact on an outcome Y . For instance, a political scientist might investigate whether exposure to a campaign mobilization message increases a voter's likelihood of turning out in an election; or an economist might study whether a direct cash transfer raises long-term income. Despite differing contexts, a central challenge unites these inquiries: *confounding variables* threaten to blur the line between correlation and causation.

Confounding arises when some factor U affects both the probability of receiving the treatment X and the outcome Y . If one does not adequately account for U , any observed relationship between X and Y might be partially or wholly attributable to U . This

paper discusses how randomization—and other design-based strategies—can break the link between confounders and treatment assignment. We also delve into the variety of *conditioning-based* methods that attempt to statistically control for confounding. In Section , we present a detailed comparison of these two overarching approaches, supported by a table that summarizes their core ideas, advantages, and common pitfalls.

A Simple DAG for Confounding

One way to visualize confounding is with a Directed Acyclic Graph (DAG). Figure 1 illustrates a minimal setup where a confounder U influences both the treatment X and the outcome Y .

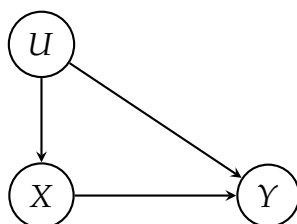


Figure 1: A simple DAG showing how a confounder U can influence both treatment X and outcome Y .

In non-experimental settings, individuals who choose (or are selected) to receive X may systematically differ from those who do not in ways captured by U . If a researcher merely regresses Y on X without accounting for U , the resulting estimate may be biased, conflating the effect of the treatment with the effect of the confounder.

Potential Outcomes Framework

The potential outcomes framework offers a coherent way to define and measure causal effects. For each individual i :

$Y_i(1)$ is the outcome if individual i is treated,

$Y_i(0)$ is the outcome if individual i is not treated.

The *Individual Treatment Effect* (ITE) for person i is

$$\text{ITE}_i = Y_i(1) - Y_i(0).$$

Because it is impossible to observe both $Y_i(1)$ and $Y_i(0)$ for the same individual, researchers typically focus on the *Average Treatment Effect (ATE)*:

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)].$$

If the assignment to treatment is truly random, the difference in average outcomes between treatment and control groups can provide an unbiased estimate of the ATE.

Why Randomization Blocks Confounding

Random assignment ensures that potential confounders, whether measured or not, are balanced (on average) across treatment and control groups. Returning to Figure 1, one can think of randomization as “cutting” the arrow from U to X . By design, who ends up receiving X no longer depends on factors like prior interest in politics or differences in wealth. As a result, any difference in outcomes $\bar{Y}_{\text{treatment}} - \bar{Y}_{\text{control}}$ can be more credibly interpreted as the causal effect of X .

Of course, real-world experiments face practical issues such as noncompliance (when participants do not adhere to their assigned treatment) and attrition (when participants drop out of the study). These issues can weaken the ideal guarantee of randomization, yet a carefully executed experiment remains the most straightforward way to isolate the effect of X from the influence of U .

Conditioning-Based vs. Design-Based Approaches

Researchers who aim to mitigate confounding typically use one of two overarching strategies: *conditioning-based* approaches or *design-based* approaches. Table 1 summarizes their main features, followed by more detailed descriptions of each category.

Conditioning-Based Methods

Conditioning-based methods, such as multiple regression, matching, or propensity-score weighting, seek to statistically control for confounders that researchers *observe and measure*. For example, one may fit a linear regression of the form

$$Y_i = \alpha + \beta X_i + \gamma_1 U_{i1} + \gamma_2 U_{i2} + \cdots + \varepsilon_i,$$

Table 1: Comparing Conditioning-Based and Design-Based Approaches

Approach	Key Idea	Typical Limitations
Conditioning-Based	Statistically adjust for observed confounders via regression, matching, or weighting. Balances measured covariates in an attempt to isolate X 's effect on Y .	Cannot address unobserved confounders; relies on correct model specification or complete measurement. Sensitive to omitted variables and parametric assumptions.
Design-Based	Leverage random or quasi-random variation in X . Examples include RCTs, DiD, IV, and RDD. Goal is to ensure or approximate that confounders are balanced by design.	Practical, ethical, or logistical constraints may prevent true randomization. Quasi-experimental methods require strong assumptions (e.g., parallel trends, valid instruments, continuity) which can be violated in real data.

where $U_{i1}, U_{i2}, \dots$ represent the observed confounders. Under the assumption that *all relevant confounders are included and correctly modeled*, β can be interpreted as the causal effect of X .

Alternatively, *matching* pairs (or groups of) treated and control units with similar observed characteristics. A common strategy is *propensity score matching*, where one estimates the probability of receiving treatment based on observed covariates, then matches treated units with control units that have similar propensity scores. If matching is well-executed and confounders are accurately measured, the distribution of U can become comparable across treatment and control sets.

The main advantage of conditioning-based approaches is that they are often straightforward to apply with standard statistical software and can flexibly incorporate many observed variables. However, they cannot handle unobserved confounders and may yield biased results if the functional form or the set of included covariates is incomplete. In practice, there is always a risk that important confounding factors were not measured or were measured incorrectly.

Design-Based Methods

By contrast, design-based methods try to *structure* the data collection or exploit naturally occurring conditions so that treatment assignment is as good as random (or at least, “as

if” random). The clearest example is a **Randomized Controlled Trial (RCT)**, in which an experimenter randomly assigns participants to receive treatment or control. This procedure, if done properly, ensures that neither observed nor unobserved confounders systematically vary with treatment assignment.

In settings where full randomization is infeasible or unethical, *quasi-experimental* designs are often used:

- **Difference-in-Differences (DiD)**: Compares the before–after changes in outcomes for a treated group and a non-treated group, assuming they would have followed parallel trends in the absence of treatment.
- **Instrumental Variables (IV)**: Relies on a variable Z (the instrument) that affects the outcome solely through its effect on X . If Z is genuinely exogenous, one can estimate the Local Average Treatment Effect (LATE).
- **Regression Discontinuity (RDD)**: Exploits a cutoff rule that assigns treatment if a running variable exceeds a threshold. Under continuity assumptions near the cutoff, comparisons just above and below the threshold approximate random assignment.

These design-based methods have the potential to address *unobserved* confounders because, in principle, the process of assignment to treatment is disconnected from individuals’ pre-existing attributes. However, strong assumptions must be met. For DiD, one needs parallel outcome trajectories for treated and control groups in the absence of treatment. For IV, the instrument must be strongly correlated with X but must not affect Y in any direct way. For RDD, individuals cannot perfectly manipulate the running variable around the cutoff, and the outcome must be sufficiently smooth. When such assumptions hold, design-based approaches can provide estimates that are robust to confounding, surpassing the limitations of conditioning-based methods.

Application: Running Shoe Performance Study

To illustrate these concepts concretely, consider a researcher investigating whether a new running shoe improves marathon performance. Let $X_i = 1$ if runner i uses the new shoe and $X_i = 0$ otherwise. The outcome Y_i is the runner’s finishing time. We examine four distinct research designs, each attempting to address confounding in different ways.

Design 1: Multivariate Regression (Conditioning-Based)

The researcher collects cross-sectional data on many runners and estimates:

$$Y_i = \alpha + \beta X_i + \gamma_1 \text{Age}_i + \gamma_2 \text{Gender}_i + \gamma_3 \text{RaceHistory}_i + \gamma_4 \text{Training}_i + \varepsilon_i,$$

where additional controls might include race difficulty and weather conditions.

Classification: This is a *conditioning-based* approach that statistically adjusts for observed confounders.

Confounders Addressed: Age, gender, race history, training volume, race difficulty, and weather—all measured variables that might influence both shoe choice and running time.

Confounders NOT Addressed: Unobserved factors such as changes in training technique that coincide with shoe adoption, sponsorship deals that provide both shoes and coaching, or innate motivation levels that drive both equipment purchases and performance improvements.

Key Assumption: All relevant confounders are observed and correctly included in the model. Formally, we assume conditional independence:

$$Y(1), Y(0) \perp X \mid \text{Age, Gender, RaceHistory, Training, etc.}$$

Design 2: Difference-in-Differences (Design-Based)

The researcher tracks the same runners competing in the same races before and after some runners switch to the new shoe. The DiD estimator is:

$$\widehat{\text{ATE}}_{\text{DiD}} = (\bar{Y}_{\text{shoe,after}} - \bar{Y}_{\text{shoe,before}}) - (\bar{Y}_{\text{no shoe,after}} - \bar{Y}_{\text{no shoe,before}}),$$

where the first term captures the change in times for runners who adopted the shoe, and the second term captures the change for runners who did not.

Classification: This is a *design-based* approach exploiting within-runner and within-race variation.

Confounders Addressed:

- *Time-invariant runner characteristics* (innate ability, body type) are differenced out by comparing the same runners over time.
- *Time-varying race characteristics* (course difficulty, weather) that affect all runners equally are differenced out by comparing runners in the same races.

Confounders NOT Addressed: Time-varying factors that differ between adopters and non-adopters, such as differential improvements in training regimens, diet changes, or increased seriousness about marathoning that correlates with shoe adoption.

Key Assumption (Parallel Trends): In the absence of the new shoe, the average change in running times would have been the same for both groups:

$$\mathbb{E}[Y(0)_{\text{after}} - Y(0)_{\text{before}} \mid \text{Shoe} = 1] = \mathbb{E}[Y(0)_{\text{after}} - Y(0)_{\text{before}} \mid \text{Shoe} = 0].$$

Design 3/4: Within-Runner Comparisons (Design-Based)

The researcher follows individual runners over time, comparing each runner's performance before and after switching to the new shoe. For runner i , the estimated effect is:

$$\widehat{\text{Effect}}_i = Y_{i,\text{after}} - Y_{i,\text{before}}.$$

Aggregating across all runners who switched gives:

$$\widehat{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n (Y_{i,\text{after}} - Y_{i,\text{before}}).$$

Classification: This is a *design-based* approach using within-runner variation to control for stable individual characteristics.

Confounders Addressed: Any time-invariant attributes of runners, such as innate running ability, biomechanics, or genetic endowment.

Confounders NOT Addressed: Time-varying factors that change simultaneously with shoe adoption:

- New training regimen implemented at the same time
- Dietary improvements
- Performance improvements leading to sponsorships that provide both new shoes and professional coaching

Key Assumption: No other factors affecting performance change at the same time as the shoe switch. Formally:

$$\mathbb{E}[Y_{i,\text{after}}(0) - Y_{i,\text{before}}(0)] = 0,$$

meaning that in the absence of the new shoe, there would be no systematic change in performance over this period.

Comparative Summary

Table 2 summarizes the four research designs, highlighting their approaches to confounding and their core assumptions.

Table 2: Comparison of Four Running Shoe Study Designs

Design	Type	Confounders Addressed	Key Assumption
Design 1: Multivariate Regression	Conditioning	Observed variables: age, gender, training, race difficulty, weather	All relevant confounders observed and correctly modeled; no omitted variable bias
Design 2: Difference-in-Differences	Design	Time-invariant runner traits (ability); time-varying race features affecting all runners equally	Parallel trends: groups would follow same trajectory absent treatment
Design 3/4: Within-Runner Comparison	Design	Time-invariant runner characteristics (innate ability, biomechanics)	No time-varying confounders correlated with shoe adoption; stable counterfactual trend

The key insight is that conditioning-based approaches (Design 1) rely on measuring and adjusting for confounders, while design-based approaches (Designs 2–4) exploit the structure of data collection to difference out certain types of confounding. Design 2 is particularly powerful because it controls for both runner-specific and race-specific factors, but it requires the strong parallel trends assumption. Designs 3 and 4 control only for time-invariant individual characteristics and are vulnerable to time-varying confounders that coincide with shoe adoption.

Conclusion

Confounding arises when a variable U influences both the probability of receiving treatment X and the outcome Y . The potential outcomes framework clarifies the difficulty of causal inference by reminding us that we can never observe both treated and untreated outcomes for the same individual. Group-level comparisons thus become a cornerstone of causal analysis, but those comparisons can be distorted by confounding.

In well-designed experiments, randomization effectively severs the pathway from U to X . As a result, differences in outcomes between treatment and control groups are more credibly interpreted as causal. Conditioning-based approaches offer statistical tools

to control for *observed* confounders, but remain vulnerable to omitted variables. Design-based methods, including RCTs and quasi-experimental techniques like DiD, IV, or RDD, seek to minimize or eliminate confounding by harnessing exogenous or quasi-exogenous variation in X .

Ultimately, the choice between conditioning-based and design-based strategies hinges on practical feasibility, ethical considerations, and data availability. Observational research often relies on conditioning-based methods due to the difficulty of enforcing randomization, yet design-based approaches can be more convincing when their assumptions are met. The running shoe example illustrates how different research designs address different types of confounders and require different identifying assumptions. Table 1 and Table 2 serve as starting points for weighing these methods, but any empirical study must carefully evaluate the plausibility of assumptions about how treatments are assigned and how outcomes are measured.

References

- Angrist, J. D., & Pischke, J.-S. (2008). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Gerber, A. S., & Green, D. P. (2012). *Field Experiments: Design, Analysis, and Interpretation*. W. W. Norton.
- Imbens, G. W., & Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press.
- Pearl, J. (2009). *Causality* (2nd ed.). Cambridge University Press.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688–701.